

WORKSHEET 2: SET THEORY

Part 1: Set notation

Problem 1 (Reading a set). Let

$$S = \{4, 1, 4, 2, 1, 3\}, \quad T = \{3, 2, 4, 1\}.$$

- (a) Rewrite S without repetitions and find $|S|$.

$$S = \boxed{} \quad |S| = \boxed{}$$

- (b) Decide whether each statement is true or false:

$$2 \in S \quad \mathbf{T} \quad \mathbf{F}$$

$$5 \in S \quad \mathbf{T} \quad \mathbf{F}$$

$$\{1, 2\} \subseteq S \quad \mathbf{T} \quad \mathbf{F}$$

$$\{1, 2\} \in S \quad \mathbf{T} \quad \mathbf{F}$$

- (c) Are S and T equal? Explain briefly.

Problem 2 (Set-builder notation).

- (a) List the elements of $\{n \in \mathbb{Z} : -3 \leq n \leq 3\}$.

- (b) Describe the set $\{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$ using set-builder notation.

- (c) List the elements of $P = \{(x, y) : x \in \{1, 2\} \text{ and } y \in \{a, b, c\}\}$.

- (d) Find $|P|$.

Problem 3 (Set operations). Let

$$U = \{1, 2, 3, \dots, 12\}, \quad A = \{n \in U : n \text{ is even}\}, \quad B = \{n \in U : n \text{ is divisible by } 3\}.$$

(a) List the elements of A and B .

$$A = \boxed{}$$

$$B = \boxed{}$$

(b) Find each of the following sets:

$$A \cap B = \boxed{}$$

$$A \cup B = \boxed{}$$

$$A \setminus B = \boxed{}$$

$$A^c = \boxed{}$$

$$B^c = \boxed{}$$

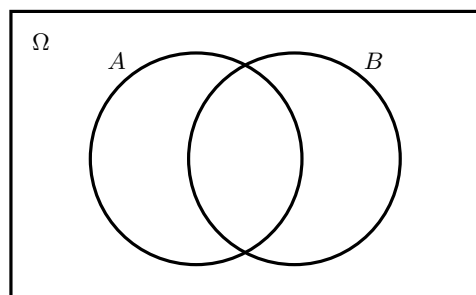
(c) Is $A \subseteq B$? Explain.

(d) Are A and B disjoint? Explain.

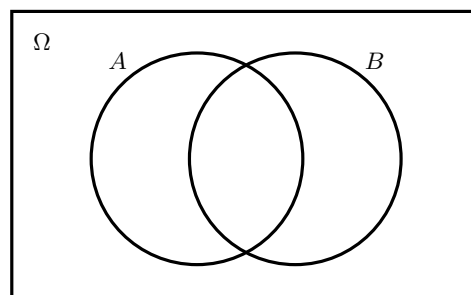
Part 2: Two important set-theoretic results

Problem 4 (De Morgan's laws). Let A, B, Ω be sets. (The symbol Ω is called Omega, and is commonly used in probability). Suppose $A, B \subseteq \Omega$ (like shown in the below two pictures.)

- (a) Shade $(A \cup B)^c$ in the first Venn diagram and $A^c \cap B^c$ in the second.



$(A \cup B)^c$



$A^c \cap B^c$

- (b) Are the shaded regions the same? Complete the identity

$$(A \cup B)^c = \boxed{}$$

Problem 5 (The inclusion–exclusion formula). The *inclusion–exclusion formula* states that if A and B are finite sets, then

$$|A \cup B| = |A| + |B| - |A \cap B|.$$

Explain why this identity holds using pictures.

Part 3: Connection to probability theory

Problem 6 (Dice sets). Suppose we roll two dice. We represent the outcome by an ordered pair (x, y) , where x is the result of the first dice and y is the result of the second. The set of all possible outcomes is

$$\Omega = \{(x, y) : x, y \in \{1, 2, 3, 4, 5, 6\}\}.$$

- (a) How many elements does Ω contain? Explain how you can count them without listing every element.

- (b) Let B be the set of outcomes for which the sum of the two dice is 7. Describe B in set-builder notation, and then list its elements.

- (c) Let C be the set of outcomes for which the two dice roll the same number. Describe C in set-builder notation, and then list its elements.

- (d) Describe the following three sets in words. What events do they represent?

$$D = \{(x, y) \in \Omega : x > y\}, \quad E = \{(x, y) \in \Omega : x = 6 \text{ or } y = 6\}, \quad F = \{(1, 1)\}.$$

- (e) In probability we use the word “event” rather than “set”. Describe the following events in set-builder notation:

- (i) The event that the dice add up to at least 10.

- (ii) The event that at least one dice shows an even number.

- (f) List the elements of the following sets

$$B \cap E = \boxed{\phantom{\{1, 2, 3, 4, 5, 6\}}}$$

$$C \setminus E = \boxed{\phantom{\{1, 2, 3, 4, 5, 6\}}}$$

- (g) Compute the following probabilities:

$$\mathbb{P}(D) = \boxed{}$$

$$\mathbb{P}(E) = \boxed{}$$

$$\mathbb{P}(F) = \boxed{}$$

$$\mathbb{P}(B \cap E) = \boxed{}$$